



## Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

### Innovative Practice Description

- **Unit / Topic:** Unit I / Regular Expression to Automata

- **Course Outcome:** CO1

- **Topic Learning Outcome:** TLO3

- **Activity Chosen:** Think Pair Share

- **Justification:**

The understanding of regular expression and its conversion to automata is essential for developing the phases of compiler. Hence the students have to think the way of writing regular expression for the given token and automata conversion. Hence this activity is chosen to make them think and communicate whether their thought process is correct or not with their neighbors.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- A Collaborative, active learning strategy, in which students work on a question posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share).
- **Think – Write-** Students were provided with questions on Regular expression and automata. The student has to think individually and write the answer in a paper and it is shown in Fig.1.
- **Pair-** After writing the answers, the students were allowed to discuss with neighbour and justify the resultant answer to each other. The answers written by the students are shown in Fig.2
- **Share -** Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team.

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO5 | PO9 | PO10 | PO12 | PSO1 | PSO2 | PSO3 |
|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1 | 3   | 3   | 2   | 1   | 2   | 1    | 2    | 2    | 1    | 1    |

(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice | PO1 | PO2 | PO3 | PO9          | PO10 | PSO1         |
|---------------------|-----|-----|-----|--------------|------|--------------|
|                     | 3   | 2   | 2   | 2            | 2    | 1            |
| Justification       | The | The | The | The students | The  | The students |

|                        |                                                                       |                                                             |                                                     |                                       |                                                          |                                                                                           |
|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|---------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>for correlation</b> | students apply the knowledge of regular expression to define patterns | students analyze the given questions and identify solutions | students find the solutions for the given question. | effectively communicate their answers | students are working as a team to provide the conclusion | apply the regular expression concepts for lexical analyzer to help in developing software |
|------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|---------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|

• **Images / Screenshot of the practice:**



**Fig1. Student's think and write answers**

B.Dharini

1) Regular expression:  
eg:-  $1^*(0+1) + (0^*1)0$

2) string:  
110, 1010, 1110, 1000  
0110, ...

3) NFA - non-deterministic finite Automata  
- has no dead state  
- more than one transition from a state.  
eg:  

```

graph LR
    q1((q1)) -- 0 --> q2((q2))
    q2 -- 1 --> q3(((q3)))
  
```

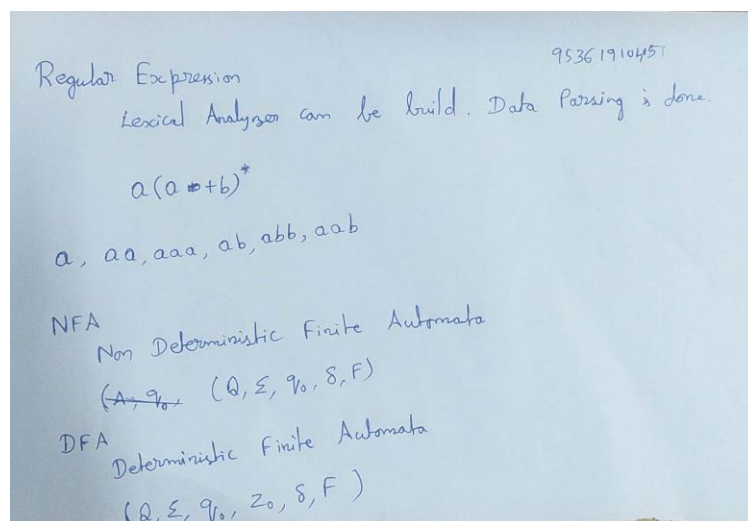
4) DFA - Deterministic Finite Automata  
- has Dead states  
- has only one transition from a state  
eg:  

```

graph LR
    q1((q1)) -- 0 --> q2((q2))
    q2 -- 1 --> q3(((q3)))
  
```

R.k.Yuvavon

1. Regular expression:  
These expression are sequence of character that specific search pattern in text.  
 $01(0+1)^*$   
 $A = \{0101, 01111, 01001, 010011\}$   
DFA:-  
- It contain 5 states.  
-  $\{Q, \Sigma, \delta, q_0, F\}$   
Q - Set of states.  
 $\Sigma$  - Set of alphabets.  
 $\delta$  - set of transition  
 $q_0$  - Initial state  
F - Final state



**Fig 2: Sample Answers**

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Some of the students write answers for the questions easily and share their ideas with others.
    - Some of them need more explanation and examples to get clarity on the regular expression.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and get more confidence, clear understanding in this topic when discussing with their peers.

- ❖ **Challenges faced in implementation:**

- The students who didn't understand the concept clearly is not able to complete it within time.
    - Making all the students to involve and present actively in this activity is a challenging.

**References:**

- ❖ <https://www.ritrjpm.ac.in/images/computer-science/DRKV-BV-Operators.pdf>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share-PJT.pdf>
- ❖ [https://www.ritrjpm.ac.in/images/computer-science/14\\_CS8392\\_TPS.pdf](https://www.ritrjpm.ac.in/images/computer-science/14_CS8392_TPS.pdf)

**Signature of Faculty Member**

**HOD**